Supplementary material:
Learning to communicate across modalities: perceptual heterogeneity in multi-agent systems

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A. Detailed Experimental Setup

This section provides the technical details of the experimental setup adopted for this work, which were briefly discussed in Section 2 of the paper. Our implementation follows Evtimova et al. (2018). However, their study addresses different research questions, focusing on visual-text communication and performance. The analyses, metrics, modalities, and embedding generation strategies differ greatly, so direct comparison of results is not meaningful.

A.1. *Agents*

Sender. The Sender is defined as:

$$A_S : O_S \times S \rightarrow S \tag{1}$$

mapping its private input $o_s \in O_S$ and the Receiver's most recent message $m^r_{t-1} \in S$ to a binary message $m^s_t \in S$ where $S = \{0, 1\}^D$ and D is the length of the message. In our implementation, o_s is an audio embedding extracted from a pretrained VGGish model, while in Evtimova et al. (2018) it was an image embedding. The Sender combines both inputs through a linear projection of each, followed by an element-wise summation. The resulting fused representation is passed through a `tanh` activation layer and subsequently projected to the predefined message dimensionality. A final sigmoid layer constrains the output values to the $[0, 1]$ range. During training, messages are generated by sampling each element from an independent Bernoulli distribution parameterised by these outputs, while at test time, they are thresholded deterministically. An overview of the Sender architecture is provided in Figure 1.

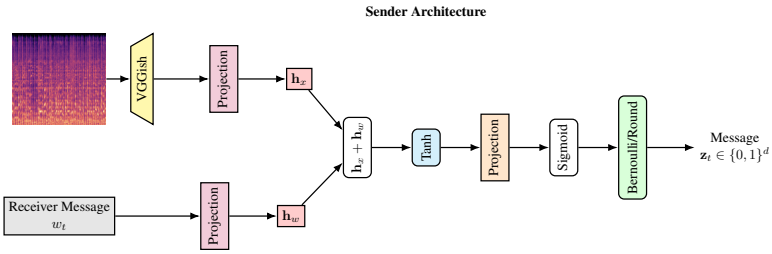


Figure 1. Architecture of the Sender agent (defined by Equation 1). The Sender encodes audio using a VGGish model and combines it with an optional message using an element-wise sum of their linear projections to produce a communication signal. During training, the message is sampled from a Bernoulli distribution to allow gradient flow. This is replaced with a deterministic thresholding strategy during evaluation.

Receiver. The Receiver is defined as

$$A_R : S \times \mathbb{R}^q \to \Xi \times O_R \times S \times \mathbb{R}^q, \tag{2}$$

where $\Xi = \{0, 1\}$ is the stop decision, O_R is the set of candidate objects, and $h_t \in \mathbb{R}^q$ represents the Receiver's internal hidden state at time step t. Given the Sender's message $m_s \in S$ and its previous hidden state h_{t-1}, the Receiver updates its state using a Gated Recurrent Unit (GRU), producing a new hidden state $h'_t \in \mathbb{R}^q$. It then computes a probability distribution over all candidate objects $o_r \in O_R$, representing its current guess about the target object. Finally, the Receiver decides whether to terminate the interaction ($s_t = 1$) or continue by generating a response message $m_r \in S$. When $s_t = 1$, the Receiver selects the most likely image from the set of candidate images, $\hat{o}_r = \arg\max_{o_r \in O_R} p(o_r = 1)$. In our setup, O_R consists of either audio embeddings obtained from a VGGish network (unimodal system) or image embeddings obtained from a VGG16 network (multimodal system). Figure 2 shows an abstracted illustration of the Receiver's architecture.

Baseline. Both agents are equipped with baseline networks B_S and B_R that estimate expected rewards, reducing the variance of policy-gradient updates. These baselines are simple feedforward networks operating on the agent's current inputs and hidden state. B_S receives as input the Receiver's message and the Sender's audio input projected through its first hidden layer (Figure 3), while B_R takes as input the Receiver's hidden state. Both networks output the predicted loss, $\hat{L} \in \mathbb{R}$.

A.2. *Training*

Agents are trained jointly for 200 epochs using RMSprop with a learning rate of 10^{-4}. During each conversation, a new set of distractors is sampled and the

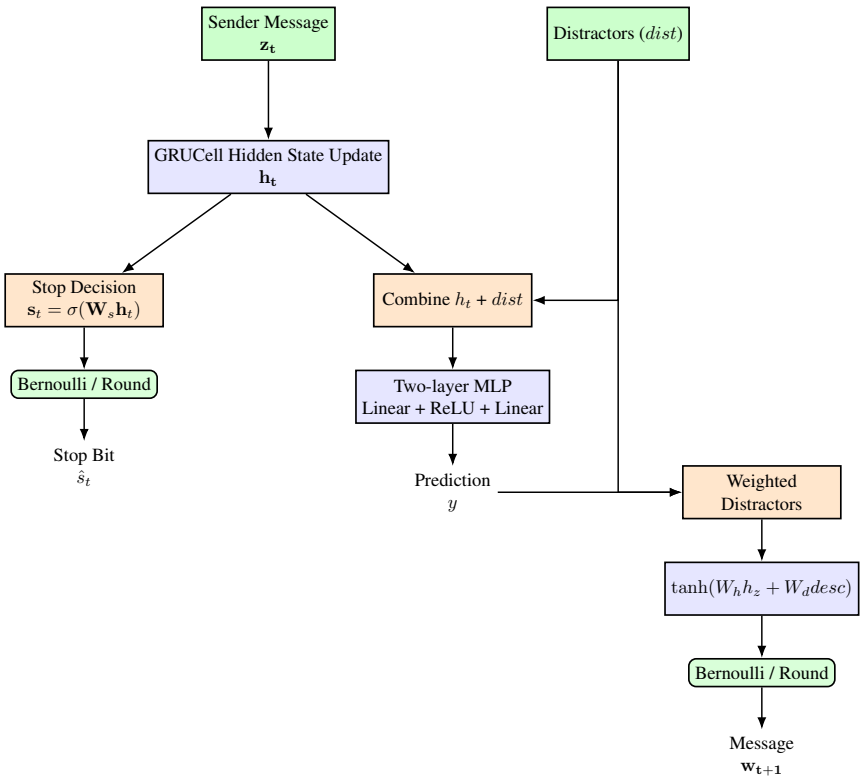


Figure 2. Abstracted architecture of the Receiver agent (defined by Equation 2). The Receiver takes as input the Sender's message and a set of distractors, updates its hidden state, computes stop decisions, generates predictions, and produces outgoing messages.

conversation proceeds (Evtimova et al. (2018) opted for a fixed distractor set) for up to $T_{\max} = 10$ steps, with the Receiver free to terminate earlier. Variable-length conversations are handled with binary masks so that only active steps contribute to the loss. Gradient clipping is applied at every update for stability. The overall objective combines supervised classification, reinforcement learning, and entropy regularisation (Equation 3):

$$L_i = L_i^c + L_i^r - \sum_{t=1}^{T} \left[\epsilon_s H(s_t) + \epsilon_m \sum_{j=1}^{d} \left(H(m_{t,j}^s) + H(m_{t,j}^r) \right) \right] \tag{3}$$

where $H(\cdot)$ denotes entropy. The classification term is the negative log-likelihood (Equation 4) of the correct candidate:

$$L_i^c = -\log p(o_r^*) \tag{4}$$

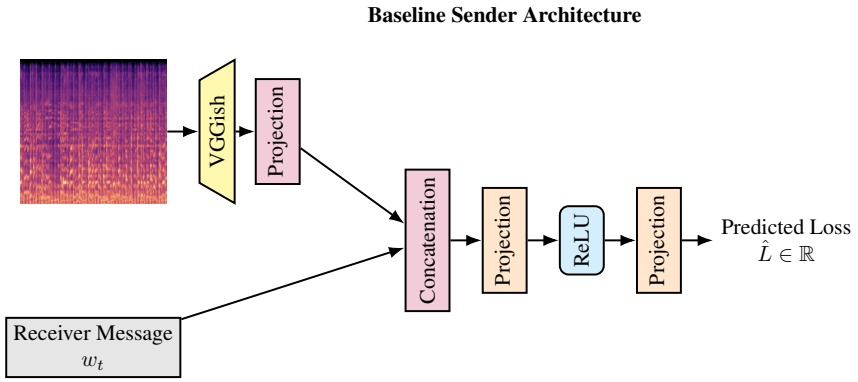


Figure 3. Architecture of the Sender Baseline agent. The Sender Baseline encodes the Sender's input using a VGGish model projected through the Sender's first hidden layer, and concatenates it with the current message. This feature vector is then passed through two Linear Layers to produce a scalar output representing the estimated loss.

The reinforcement learning term uses REINFORCE (Williams, 1992) with baseline expected loss estimation subtraction (Equation 5):

$$L_i^r = \sum_{t=1}^{T}(R - B_S)\sum_{j=1}^{d}\log p(m_{t,j}^s) + \sum_{t=1}^{T}(R - B_R)\Big[\log p(s_t) + \sum_{j=1}^{d}\log p(m_{t,j}^r)\Big] \tag{5}$$

where R is the reward given by the Receiver's classification, and B_S, B_R are baseline predictions for Sender and Receiver. The baseline networks are trained with mean squared error.

A.3. *Data Collection and Preprocessing*

Synthetic Dataset (Shapes World). To establish a controlled proof of concept, we construct a synthetic multimodal dataset consisting of six abstract classes: *circle*, *square*, *triangle*, *heart*, *star*, and *hexagon*. The dataset is designed to provide a structured and flexible environment for exploring emergent communication under noise-controlled and systematically variable conditions. A total of 2,400 images are generated using a Python-based image synthesis pipeline, with 400 samples per class. Each image depicts one of the six shapes rendered on a blank white background. Randomisation is applied to encourage generalisation while maintaining distinct inter-class boundaries. Specifically, for each instance, each shape is assigned a:

- **Colour:** random RGB, uniformly sampled from $[0, 255]$ for each channel.
- **Shape Size:** dimensions vary between 50 and 150 pixels.

- **Rotation:** random angle between $0°$ and $360°$ is applied to each shape.
- **Position:** random centre coordinates such that the entire figure remains within the 256×256 pixel canvas.

In parallel, a corresponding audio dataset comprising 1,200 clips (200 per class) is generated using `NumPy` and `SciPy`. Each clip represents one of the six shape classes through a class-specific waveform pattern with controlled stochasticity. To simulate acoustic variation, several signal attributes are randomised:

- **Frequency:** sampled uniformly from 200Hz to 800Hz.
- **Amplitude:** randomly selected from $[0.3, 0.9]$ (introduces loudness variation).
- **Gaussian Noise:** Zero-mean Gaussian noise ($\sigma = 0.02$) is added.

For perturbation experiments, all other attributes are held constant while the frequency is separated into predefined ranges – specifically, 200–300Hz, 400–500Hz, and 600–800Hz.

Environmental Dataset. To evaluate performance under more realistic and noisy conditions, we construct a multimodal environmental dataset by pairing natural images with environmental sounds. Six visual classes are selected from the CIFAR-100 dataset (Krizhevsky, 2009) (*e.g.*, *wolf*, *lion*, *pickup truck*), each containing 600 images resized and relabeled for consistency. Corresponding acoustic categories are drawn from the UrbanSound8K (Salamon et al., 2014) and ESC-50 datasets (Piczak, 2015) (*e.g.*, *dog bark*, *siren*, *car horn*), also balanced to 600 samples per class. To address data scarcity, audio samples are augmented using the `audiomentations` (Bilenko, 2021) library with operations such as Gaussian noise addition, pitch shifting, time stretching, and gain adjustment. While some cross-modal mappings are imperfect (e.g., pairing *wolf* images with "dog bark" sounds), precise semantic alignment is not required for our purposes. The visual and auditory features originate from fundamentally different embedding spaces, and no inherent correlation exists between them prior to interaction. Any relationship between the modalities emerges only as the agents develop it. What matters for the analysis is whether agents can construct protocols that bridge abstract concepts based on their own representations rather than real-world semantics.

Audio Preprocessing. Every 0.96 seconds of audio inputs are transformed into 128-dimensional embeddings using a pretrained VGGish network (Taylor, 2019). These embeddings are then concatenated, and dimensionality is reduced to 128 dimensions using principal component analysis (PCA) (Jolliffe, 2002).

Image Preprocessing. Image inputs are embedded using VGG16 (Simonyan & Zisserman, 2015). For the synthetic Shapes World dataset, the VGG16 model is fine-tuned on the generated shapes. For the environmental dataset, we use a pre-trained VGG16 without fine-tuning. The classification head is replaced with a 128-dimensional embedding layer, producing fixed-length feature vectors for each image. Embeddings are zero-mean normalised, as this was found to significantly improve the accuracy of the systems.

B. Extended Results

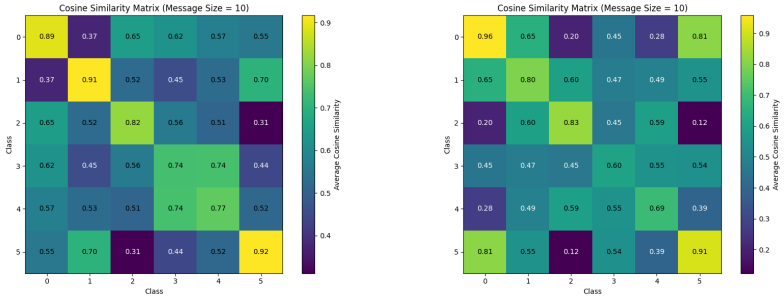
This section presents extended experimental results, initially introduced in Section 3 of the main paper.

B.1. *Perceptual Misalignment and Efficiency*

This section expands on the experiments discussed in Section 3.1 of the paper. The impact of message length on system performance metrics is shown in Table 1 of the paper. Contrary to expectations, Receiver entropy remains largely unaffected by message size, except when communication becomes overly constrained, where entropy begins to increase. Additionally, the multimodal system reaches its communication limits at longer message lengths compared to the unimodal system, indicating a greater sensitivity to restricted communication capacity. Receiver entropy is also consistently lower in the unimodal system. These patterns were visible across both synthetic and environmental datasets.

B.2. *Class Consistency of Sender-Receiver Protocols*

This section presents further details on the experiments described in Section 3.2. Figure 4 presents a matrix of average cosine similarity between the Sender's first messages for instances of the same class. Both the unimodal and multimodal systems converge toward consistent, class-similar messages. Classes with higher cross-class confusion, such as Classes 3 and 4, exhibit higher cross-class message similarity, indicating overlap in their communication representations. However, within-class similarity remains stronger in the multimodal setting for these same classes, suggesting improved message distinction and clearer separation between classes compared to the unimodal system, likely due to interaction with the visual modality.



(a) Unimodal (b) Multimodal

Figure 4. Cosine similarity matrices of Sender message embeddings for the unimodal and multimodal systems at message length 10. Sender messages within classes remain consistent in both unimodal and multimodal setups.

We conducted perturbation trials across 10 runs, randomly flipping selected bits and recording the system's average classification accuracy and variance. Across all classes, the Sender-Receiver message pairs shared at most a single agreed bit. To examine the influence of these bits, we systematically flipped them in the Sender's first message and measured the Receiver's resulting classification performance. As shown in Figure 5, flipping these bits had a negligible effect on accuracy in most classes, except for the circle class, where performance declined slightly. This suggests that while some agreed bits may occasionally contribute to message interpretation, they are more likely to represent constant elements within the Sender's encoding rather than shared, semantically meaningful components of a common communication protocol.

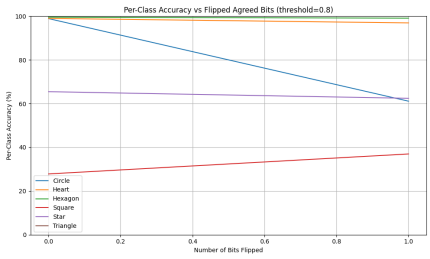
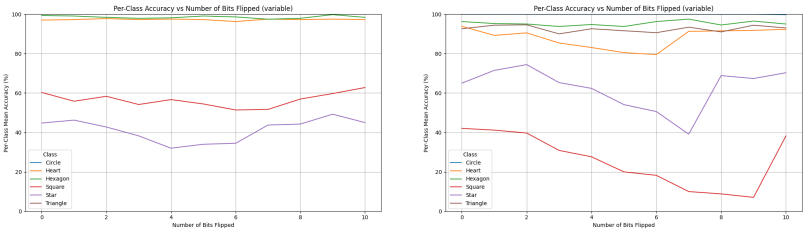


Figure 5. Effect of flipping agreed bits on Receiver accuracy. Results show minimal effect except for the circle class.

B.3. *Grounding and Semantics*

This section presents further information on the experiments detailed in Section 3.3 of the paper. Based on prior bit-usage analysis, bits are classified as

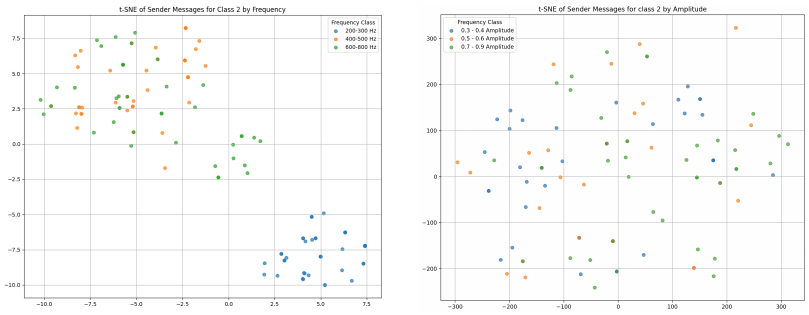
constant (*i.e.*, those that are active (1) more than 90% or inactive (0) less than 10% of the time), or *variable* (*i.e.*, those that fluctuate between 0 and 1). Subsets of these bits are flipped, and the resulting classification accuracy and variance are measured. Conversations are restricted to a single exchange, since prior results show that the first message already conveys sufficient information for high task accuracy. Figure 6 shows the per-class accuracy when flipping variable bits. The effect of flipping such bits notably has a larger effect in the unimodal compared to the multimodal system, indicating that these bits encode the Sender's fine-grained perceptual information that the multimodal Receiver, operating on a different modality, is less sensitive to.



(a) Multimodal (b) Unimodal

Figure 6. Per-class accuracy when flipping variable (fluctuating) bits. Variable bit flips have minimal impact compared to constant bit flips, with a notably larger effect in the unimodal setup.

As illustrated for Class 2 (heart class) in Figure 7, messages describing lower-frequency audio inputs tend to form distinct clusters, separate from those representing higher-frequency inputs. This pattern suggests that frequency influences the structure of the Sender's message space. In contrast, variations in amplitude do not produce such clustering, indicating that the communication protocol is more sensitive to frequency-based differences than to changes in signal intensity.



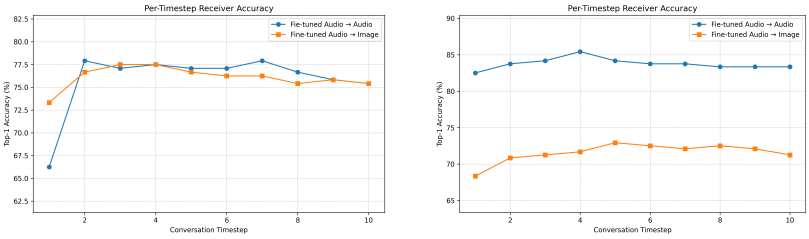
(a) Varying Frequency (b) Varying Amplitude

Figure 7. t-SNE plots of Class 2 (heart class) messages across varying frequencies and amplitudes. Messages corresponding to lower frequencies exhibit a consistent underlying pattern, while variations in amplitude show minimal impact on message structure.

B.4. *Interoperability Across Modalities*

This experiment, touched upon in Section 3.4 of the main paper, investigates whether communication protocols learned in one modality pairing can generalise to another, and how fine-tuning affects this transfer. Specifically, we examine the adaptability of an audio-based Sender trained in a multimodal setting (communicating with a visual Receiver) when it is later paired with an audio-based Receiver trained in a unimodal setup (communicating through audio only). The audio-based Sender, originally trained to communicate with an image-based Receiver, is subsequently fine-tuned with an audio-based Receiver. This setup tests the Sender's ability to adapt its communication strategy when its partner's modality changes. After fine-tuning, the Sender is evaluated with both Receivers to assess (i) how effectively it communicates within the new audio-audio pairing and (ii) whether it retains compatibility with its original image-based partner.

Our findings show that with no or minimal fine-tuning, the original multimodal pairing maintains strong performance, whereas the newly paired audio Receiver performs poorly. However, as fine-tuning progresses, the system learns to communicate effectively with both audio-based and image-based Receivers, achieving transfer with only a slight reduction in overall accuracy relative to the original multimodal setup (Table 2, main paper). Analysis of accuracy across conversation timesteps (Figure 8) reveals that after 15 epochs, the original multimodal system remains stable, with slight increases and decreases in accuracy throughout the conversation timesteps, while the audio-based system exhibits a sharp accuracy increase immediately after the first message exchange. This suggests that this first interaction might help the Sender identify its partner and adjust its communication protocol accordingly. After 100 epochs, both systems show stable communication throughout the exchange.



(a) 15 epochs

(b) 100 epochs

Figure 8. System accuracy across conversation timesteps for a multimodal trained image-based and unimodal trained audio-based Receivers. The audio Receiver shows a sharp accuracy increase after the first exchange at 15 epochs of fine-tuning, while the multimodal Receiver remains stable. Both become consistent after extended fine-tuning.

References

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